

Four Dimensional Embedding Increases Accuracy in Body Region Matching

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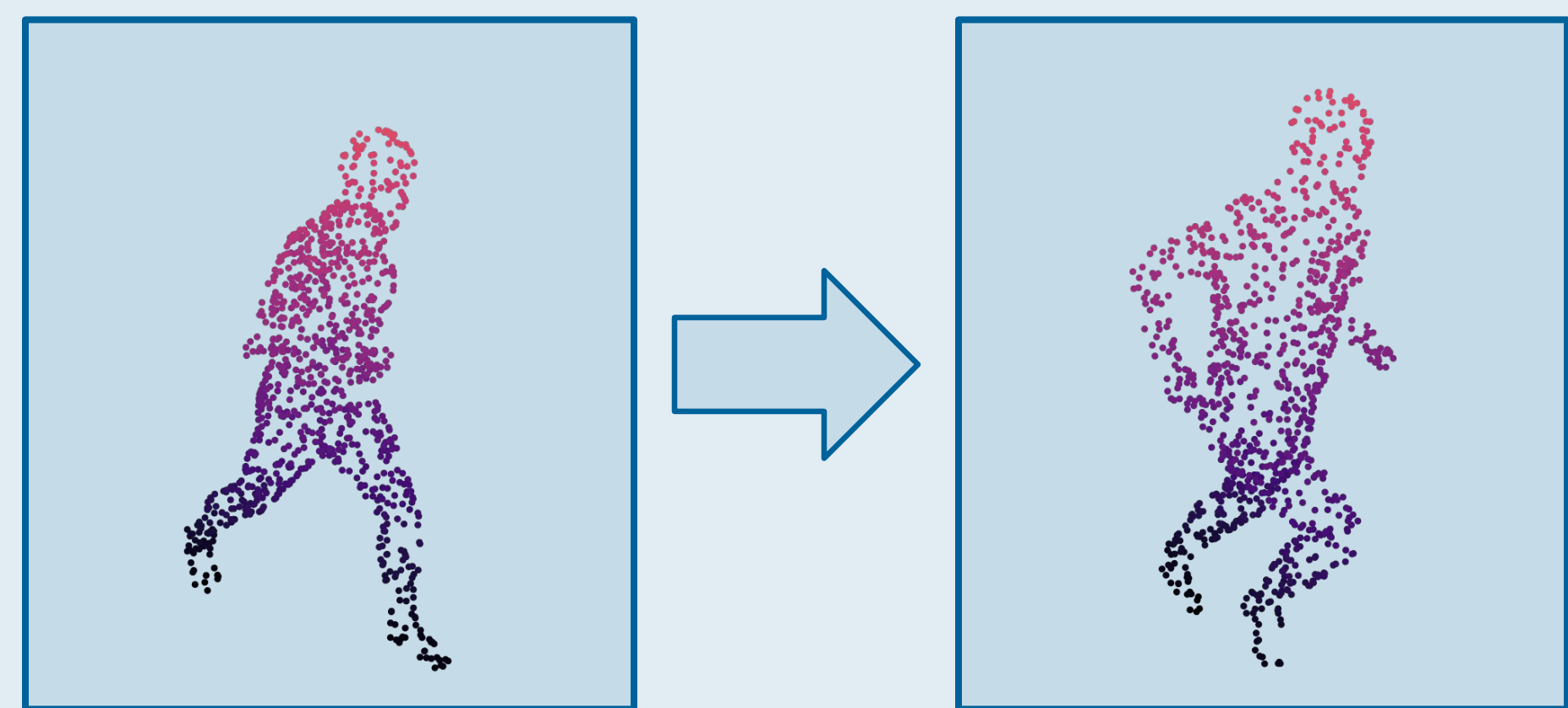
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QR Code:
(github)



Background

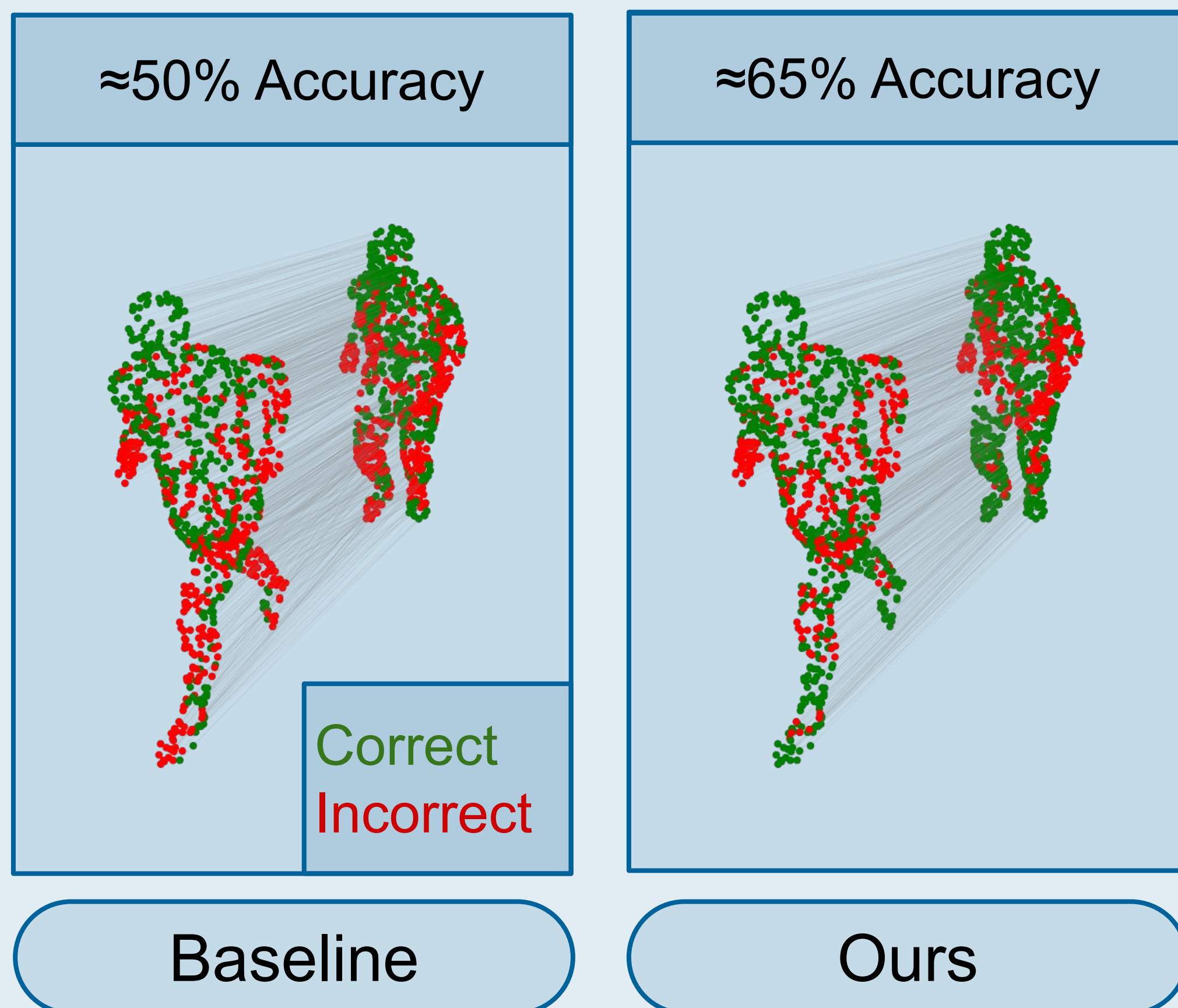
- High spatiotemporal resolution point clouds common in applications
- Matching similar regions across point clouds for action recognition, skeleton construction
- Optimal transport for one-to-one point cloud mapping; struggles with human bodies
- Key failure: confusion between right vs. left legs



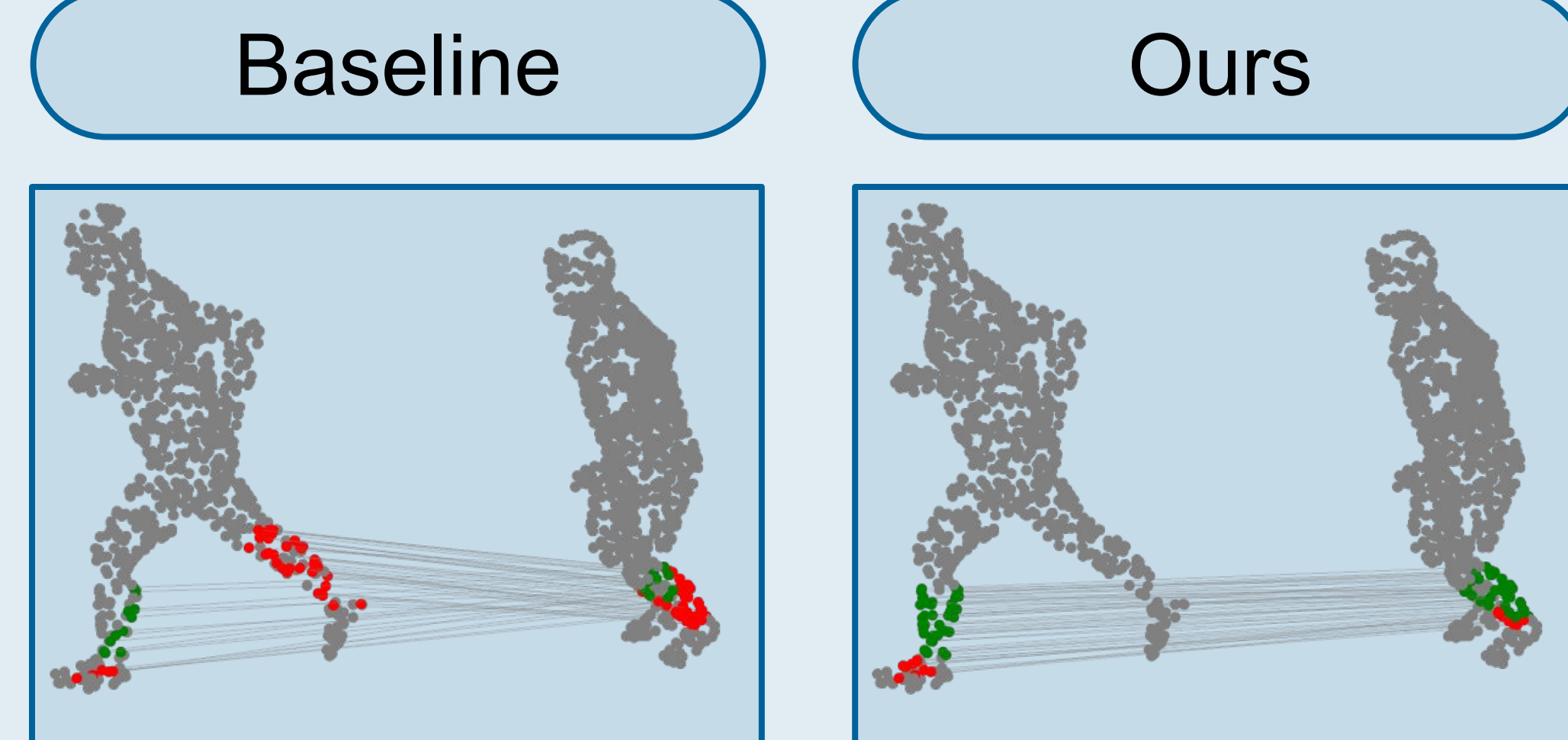
Research Question

How can we improve an optimal transport-based point matching for body regions?

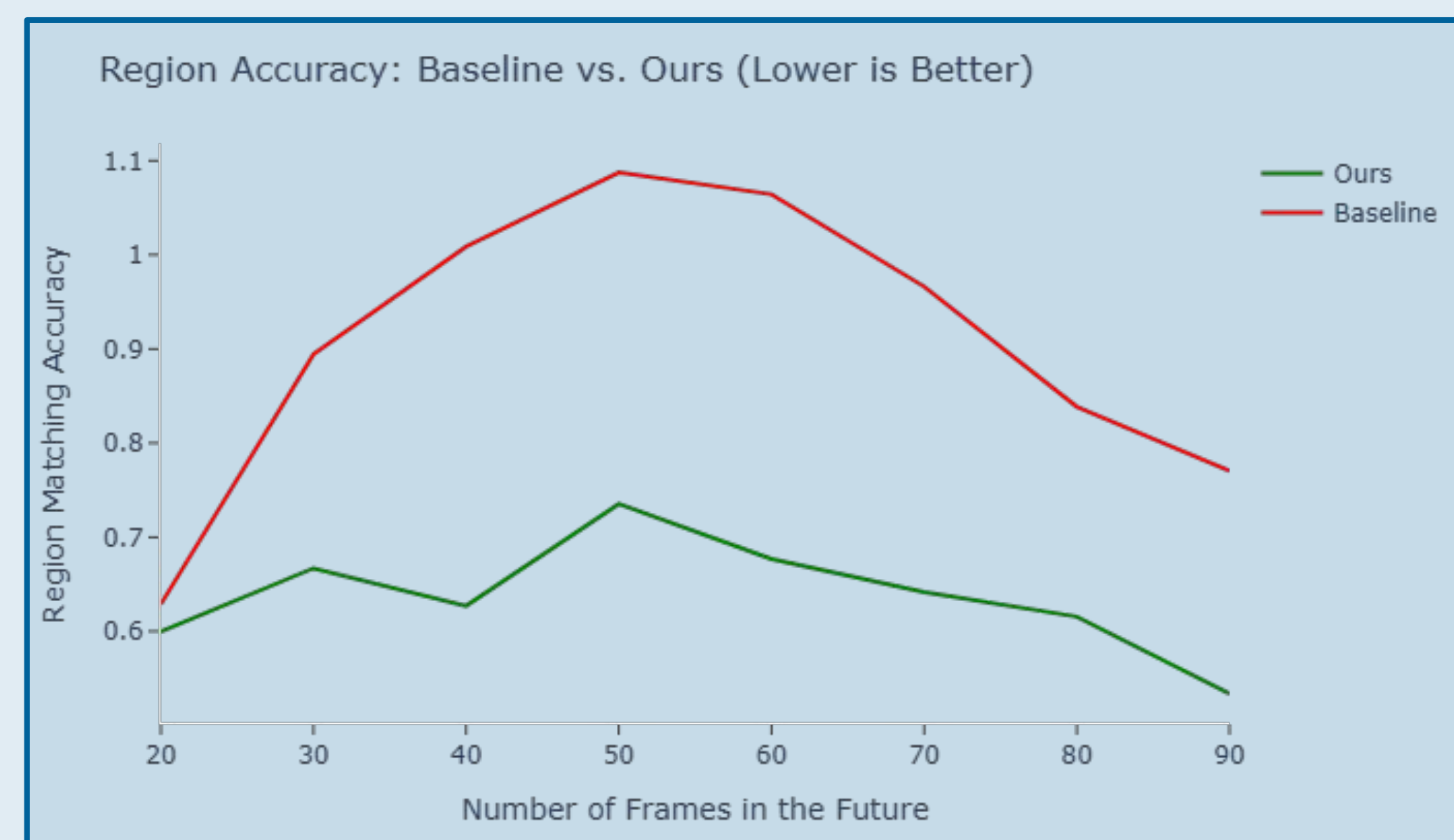
Results



- Improved leg matching
- Shin doesn't split

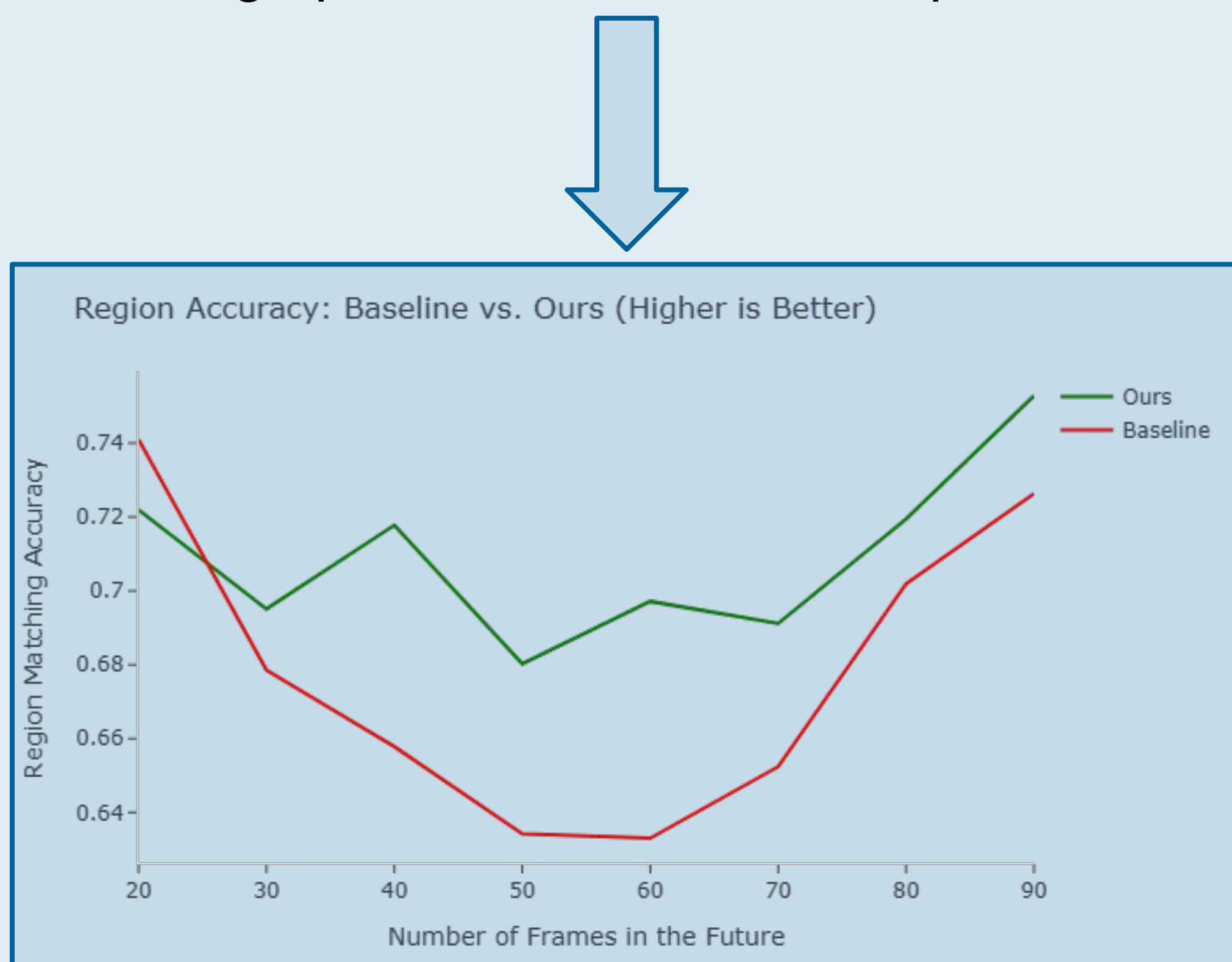


- Huge improvement overall
- Especially lower body

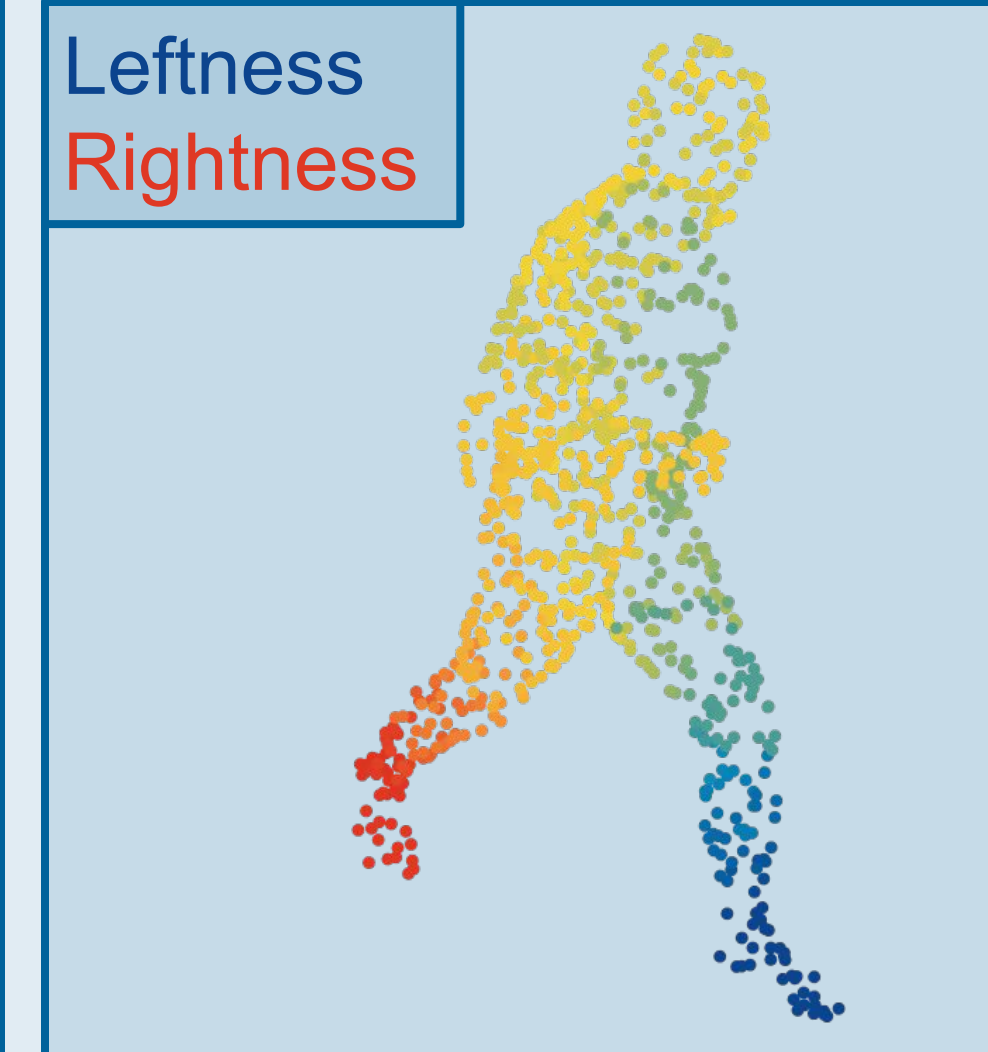
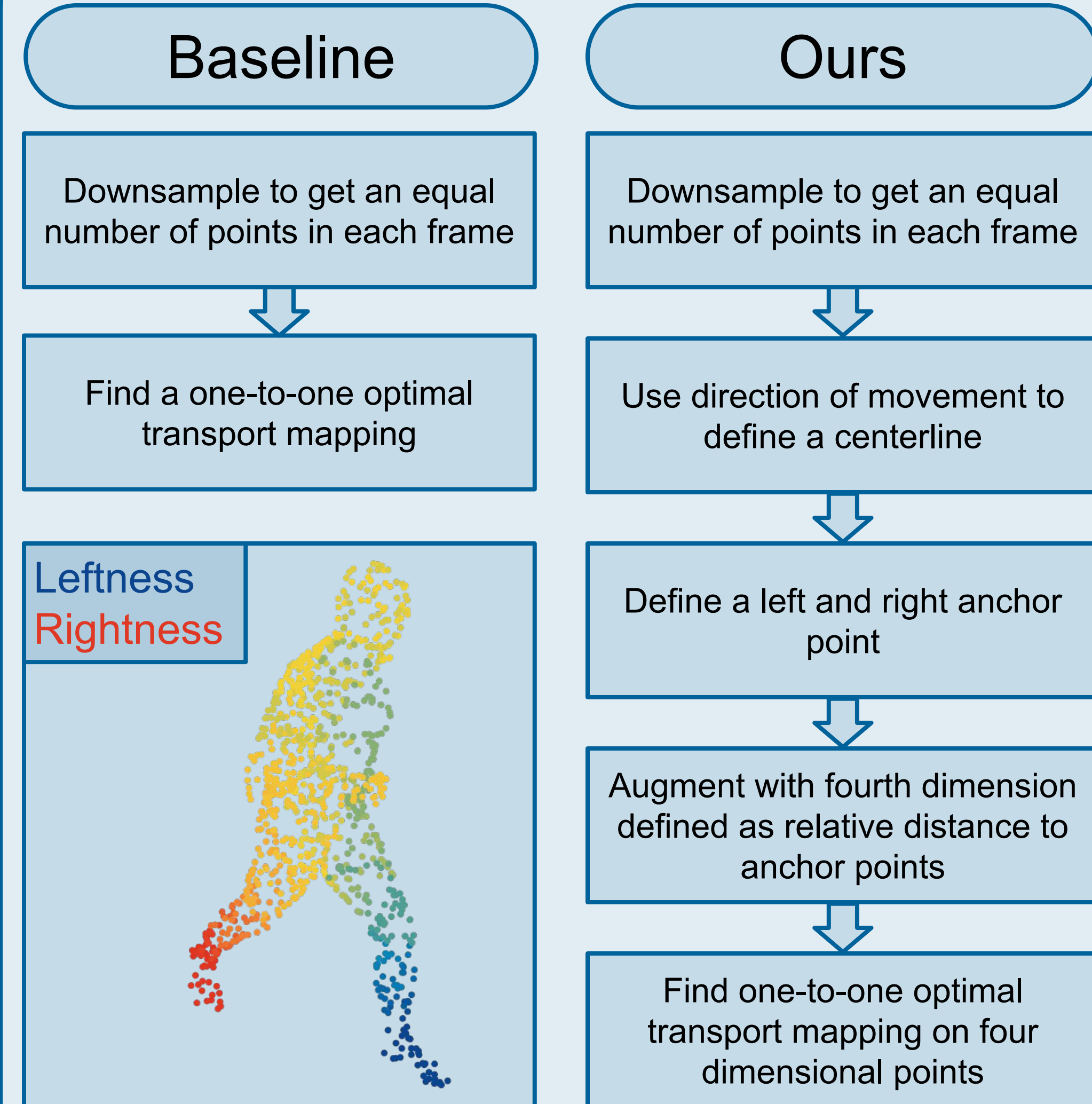


- Much closer on average, even on mismatches

- Less accuracy decay for further point clouds
- Average performance across 20 poses

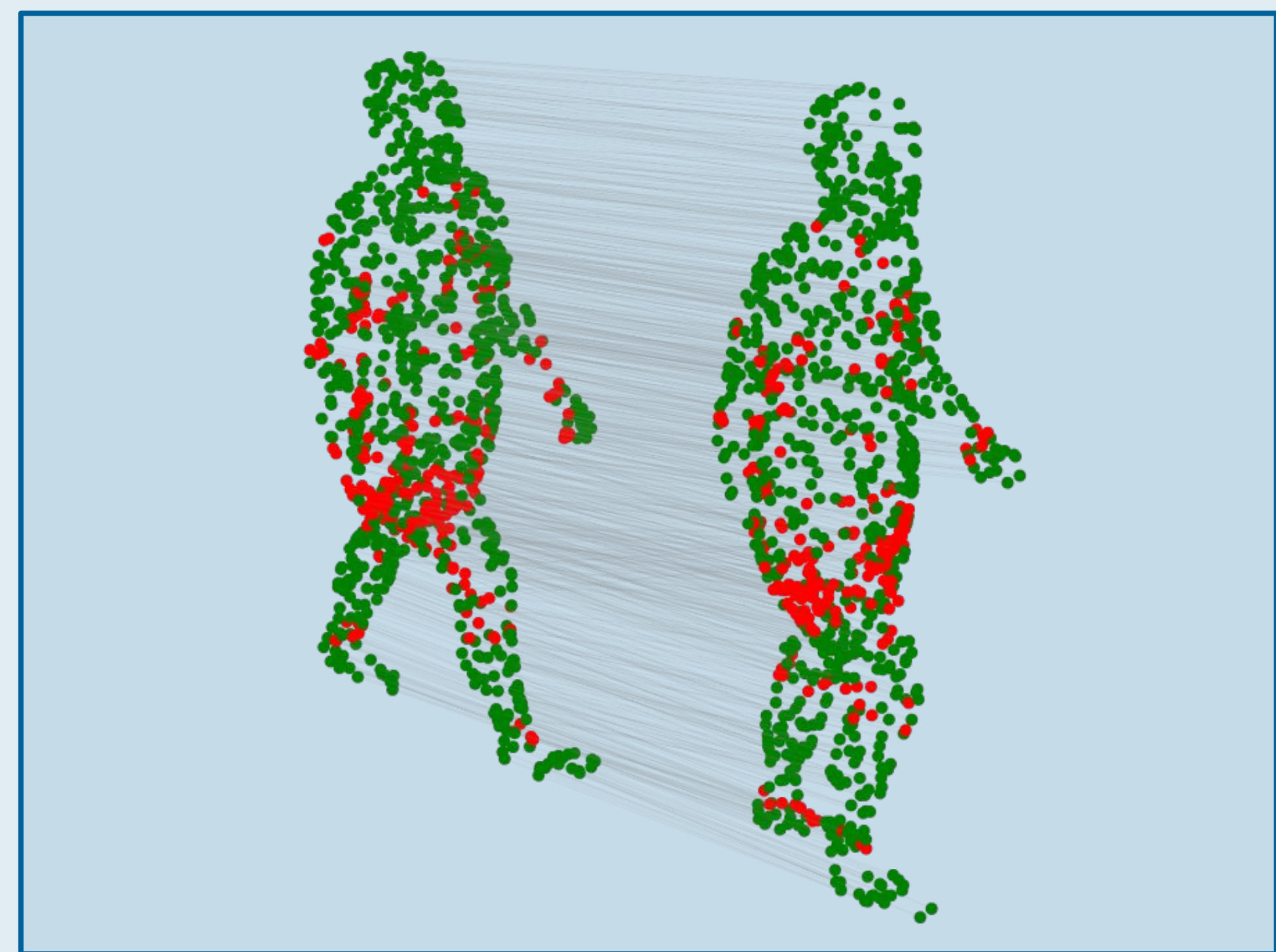


Methods



Conclusions

- Disambiguating left vs. right extremities improves region-preserving point cloud mapping
- Higher accuracy overall
- Less severe errors when mismatches occur



Data

- Point clouds sampled from AMASS human body meshes
- 1,000 surface points per frame
- Walking and running sequences (≈120 fps)
- Mesh-derived body regions used as ground truth
- 16 manually defined regions

